Configure IPv6 Addressing

1) Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : lan
   IPv6 Address. . . . . . . . . . . : 2409:40c1:5017:b8c8:bc01:5ceb:1ce7:c87a
   Temporary IPv6 Address. . . . . . : 2409:40c1:5017:b8c8:1854:8dc0:c1f2:9826
   Link-local IPv6 Address . . . . . : fe80::da83:50c3:9b3e:e0a9%10
   IPv4 Address. . . . . . . . . . . : 192.168.31.104
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::9a87:4cff:fe58:8e65%10
```

2) Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

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netsh interface ipv6 add address "Ethernet" fd00:1234:5678:1::10

ping -6 fd00:1234:5678:1::20

3) Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

Steps:

1. Checked current IPv6 addresses using ipconfig.
2. Observed temporary IPv6 address: 2001:db8:1:1:abcd:1234:5678:1111
3. Restarted the network adapter.
4. Checked again.
5. New temporary IPv6 address: 2001:db8:1:1:abcd:5678:9999:2222

4) Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

```
restaurants = {
  "Pizza Hut": "fd00:1234:5678:1::11",
  "Dominos": "fd00:1234:5678:1::12",
  "KFC": "fd00:1234:5678:1::13",
  "Burger King": "fd00:1234:5678:1::14",
  "Subway": "fd00:1234:5678:1::15"
}

search = input("Enter restaurant name: ").lower()

found = False

for name, ip in restaurants.items():
  if search in name.lower():
    print("Restaurant:", name)
    print("IPv6 Address:", ip)
    found = True

if not found:
  print("Restaurant not found.")
```

output:

```
Enter restaurant name: bur

Restaurant: Burger King
IPv6 Address: fd00:1234:5678:1::14
```

5) Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

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2001:db8:100::/48

Department	**Subnet**
IT	2001:db8:100:1::/64
HR	2001:db8:100:2::/64
Finance	2001:db8:100:3::/64

Assign your PC:
2001:db8:100:1::25/64

Gateway:
2001:db8:100:1::1

Verified using: ipconfig (or) ip -6 addr